

Formal proof systems for propositional logic

Hilbert proof system

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Formal proofs are at the core of modern logic

- The modern notion of **symbolic formal proof** was developed in the 20th century by logicians and mathematicians such as Russell, Frege and Hilbert.
- The benefit of symbolic formal proofs is that they are based on **pure syntax, i.e., symbolic manipulation**: *a precisely defined symbolic formalism with procedures for transforming statements into other statements, based solely on their **form**.*
- **No intuition or interpretation is needed**, merely applications of agreed upon rules to a set of agreed upon formulas.

The notion of formal system

Intuitively, a **formal system** consists of a language over some alphabet of symbols together with initial strings and production rules that are used to generate some of the strings in the language. More precisely,

Formal system

A formal system has the following components:

- An alphabet of symbols.
- A **syntax** that defines which sequences of symbols are in the language of our formal system.
- A decidable set of **initial sequences**.
- A finite set of **production rules** from which the set of **conclusions** of the system is generated (each conclusion is generated through a finite set of application of the production rules on both the initial sequences and the already generated conclusions).

Proof system

Formal proof system

A symbolic formal proof system (or simply **proof system**) Σ for propositional logic is constituted by:

- The set of **axioms** of Σ , i.e., a set of propositional formulas such that there is an effective procedure to decide if a given formula is an axiom.
- The (finite) set $R_1, \dots, R_n$ of **inference rules** of Σ , where each inference rule R_i is a relation that associates to a set of j_i formulas $A_1, \dots, A_{j_i}$, one formula A that is called the **direct consequence of $A_1, \dots, A_{j_i}$ through R_i** .

A formula A that is a direct consequence of $A_1, \dots, A_j$ through R is said to be **derived from $A_1, \dots, A_j$ by R** . When we want to describe an inference rule R applied to $A_1, \dots, A_j$ to derive A , we write:

$$R \quad \frac{A_1, \dots, A_j}{A}$$

Proof, provability, theorem

Proof

A **proof in Σ** is a finite sequence of formulas $\phi_1, \dots, \phi_n$ such that each ϕ_k is either

- an axiom of Σ , or
- a direct consequence of a set of previous formulas in the sequence through an inference rule of Σ .

Proof of ϕ

A **proof of ϕ in Σ** is a proof $\phi_1, \dots, \phi_n$ in Σ such that $\phi_n = \phi$.

Theorem

A formula ϕ is **provable in Σ** if there is a proof of ϕ in Σ , and in this case we write $\vdash_{\Sigma} \phi$ (or simply $\vdash \phi$ when Σ is understood), and we also say that ϕ is a **theorem** in (or of) Σ .

Proof from hypotheses

Proof from hypothesis

A **proof of ϕ in Σ from a set of hypothesis (or assumptions) Γ** is a sequence of formulas

$\phi_1, \dots, \phi_n$, with $\phi_n = \phi$, such that each ϕ_k is either

- an axiom of Σ , or
- an **hypothesis** in Γ , or
- a direct consequence of a set of previous formulas in the sequence through an inference rule of Σ .

Inference or derivation

We say that **ϕ is inferred (or derived) from Γ in Σ** , written $\Gamma \vdash_{\Sigma} \phi$ (or simply $\Gamma \vdash \phi$ when Σ is understood), if there is a proof of ϕ in Σ from Γ .

Consistency

We say that a set Γ of formulas is **Σ -consistent** if no formula ϕ exists such that both $\Gamma \vdash_{\Sigma} \phi$ and $\Gamma \vdash_{\Sigma} \neg\phi$, otherwise is **Σ -inconsistent**.

Properties of a proof system

Analogously to logical implication, it is interesting to check whether the following properties hold for a certain $\vdash_\Sigma$:

- **Reflexivity** (If $A \in \Gamma$, then $\Gamma \vdash_\Sigma A$)
- **Ex falso sequitur quodlibet** (If Γ is unsatisfiable, then $\Gamma \vdash_\Sigma A$ for all A)
- **Monotonicity** (If $\Gamma \vdash_\Sigma A$ then $\Gamma \cup \Delta \vdash_\Sigma A$)
- **Cut** (If $\Gamma \vdash_\Sigma A$ and $\Delta \cup \{A\} \vdash_\Sigma B$ then $\Gamma \cup \Delta \vdash_\Sigma B$)
- **Compactness** (If $\Gamma \vdash_\Sigma A$, then there is a finite subset $\Gamma_0 \subseteq \Gamma$, such that $\Gamma_0 \vdash_\Sigma A$)
- **Deduction principle** ($\Gamma \cup \{A\} \vdash_\Sigma B$ if and only if $\Gamma \vdash_\Sigma A \rightarrow B$)
- **Refutation principle** ($\Gamma \vdash_\Sigma A$ if and only if $\Gamma \cup \{\neg A\}$ is inconsistent)
- **Consistency**: $\vdash_\Sigma$ is consistent if there is no formula A such that both $\vdash_\Sigma A$ and $\vdash_\Sigma \neg A$

Hilbert formal proof system for propositional logic

We study the **Hilbert proof system**, that we denote by $\mathcal{H}$. In what follows, ϕ and ψ denote formulas.

Axioms of $\mathcal{H}$

$$\mathbf{A1} \quad \phi \rightarrow (\psi \rightarrow \phi)$$

$$\mathbf{A2} \quad (\phi \rightarrow (\psi \rightarrow \theta)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \theta))$$

$$\mathbf{A3} \quad (\neg\psi \rightarrow \neg\phi) \rightarrow ((\neg\psi \rightarrow \phi) \rightarrow \psi)$$

Inference rule(s) of $\mathcal{H}$

$$\mathbf{MP} \quad \frac{\phi, \phi \rightarrow \psi}{\psi}$$

Why there are no axioms for $\wedge$ and $\vee$ and $\equiv$?

Formulas with connectives $\wedge$ and $\vee$ are rewritten into equivalent formulas containing only $\rightarrow$ and $\neg$.

$$A \wedge B \equiv \neg(A \rightarrow \neg B)$$

$$A \vee B \equiv \neg A \rightarrow B$$

$$A \equiv B \equiv \neg((A \rightarrow B) \rightarrow \neg(B \rightarrow A))$$

Example of proof in $\mathcal{H}$

Example (Proof of $A \rightarrow A$ in $\mathcal{H}$)

1. $A1$ $A \rightarrow ((A \rightarrow A) \rightarrow A)$
2. $A2$ $(A \rightarrow ((A \rightarrow A) \rightarrow A)) \rightarrow ((A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A))$
3. $MP(1, 2)$ $(A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$
4. $A1$ $(A \rightarrow (A \rightarrow A))$
5. $MP(4, 3)$ $A \rightarrow A$

Example of inference in $\mathcal{H}$

Example (Inference of B from $\{A, \neg A\}$ in $\mathcal{H}$)

We show that $\{A, \neg A\} \vdash_{\mathcal{H}} B$.

- | | | |
|----|-------------------|--|
| 1. | <i>hypothesis</i> | A |
| 2. | $A1$ | $A \rightarrow (\neg B \rightarrow A)$ |
| 3. | $MP(1, 2)$ | $\neg B \rightarrow A$ |
| 4. | <i>hypothesis</i> | $\neg A$ |
| 5. | $A1$ | $\neg A \rightarrow (\neg B \rightarrow \neg A)$ |
| 6. | $MP(4, 5)$ | $\neg B \rightarrow \neg A$ |
| 7. | $A3$ | $(\neg B \rightarrow \neg A) \rightarrow ((\neg B \rightarrow A) \rightarrow B)$ |
| 8. | $MP(6, 7)$ | $(\neg B \rightarrow A) \rightarrow B$ |
| 9. | $MP(3, 8)$ | B |

Exercise on $\mathcal{H}$

Exercise on the Hilbert proof system

Prove that a set of formulas Γ is $\mathcal{H}$ -inconsistent if and only if $\Gamma \vdash_{\mathcal{H}} A$ for all formulas A .

Properties of $\mathcal{H}$

It is interesting to check whether the following property hold for $\vdash_{\mathcal{H}}$:

- **Reflexivity** (If $A \in \Gamma$, then $\Gamma \vdash_{\mathcal{H}} A$)
- **Ex falso sequitur quodlibet** (If Γ is unsatisfiable, then $\Gamma \vdash_{\Sigma} A$ for all A)
- **Monotonicity** (If $\Gamma \vdash_{\mathcal{H}} A$ then $\Gamma \cup \Delta \vdash_{\mathcal{H}} A$)
- **Cut** (If $\Gamma \vdash_{\mathcal{H}} A$ and $\Delta \cup \{A\} \vdash_{\mathcal{H}} B$ then $\Gamma \cup \Delta \vdash_{\mathcal{H}} B$)
- **Compactness** (If $\Gamma \vdash_{\mathcal{H}} A$, then there is a finite subset $\Gamma_0 \subseteq \Gamma$, such that $\Gamma_0 \vdash_{\mathcal{H}} A$)
- **Deduction principle** ($\Gamma \cup \{A\} \vdash_{\mathcal{H}} B$ if and only if $\Gamma \vdash_{\mathcal{H}} A \rightarrow B$)
- **Refutation principle** ($\Gamma \vdash_{\mathcal{H}} A$ if and only if $\Gamma \cup \{\neg A\}$ is $\mathcal{H}$ -inconsistent)
- **Consistency**: $\vdash_{\mathcal{H}}$ is consistent (i.e., there is no formula A such that both $\vdash_{\mathcal{H}} A$ and $\vdash_{\mathcal{H}} \neg A$)

Exercises on $\mathcal{H}$

Exercise: deduction theorem in the Hilbert proof system

Prove that the deduction theorem holds for the Hilbert proof system.

Exercise: cut principle in the Hilbert proof system

Prove that the cut principle holds for the Hilbert proof system.

Difference between logical implication and inference

- Logical implication

- ▶ **Definition:** a formula A is **logically implied** by a set of formulas Γ if every model of Γ is also a model of A
- ▶ **Notation:** $\Gamma \models A$
- ▶ **Nature:** based on semantics, it characterizes the **model theoretic view of logic**

- Inference

- ▶ **Definition:** a formula A is **inferred** by a set of formulas Γ in the proof system Σ if there is a proof of A in Σ from Γ .
- ▶ **Notation:** $\Gamma \vdash_{\Sigma} A$, or simply $\Gamma \vdash A$
- ▶ **Nature:** based on syntax (Σ is concerned with the **form** of axioms and inference rules), it characterizes the **proof theoretic view of logic**

Which is the relationship between the two?

Soundness and completeness of a proof system

Two very important properties of a proof system derive from the relationship between $\vdash_{\Sigma}$ and $\models$

Soundness

A proof system Σ is **sound** if for every formula A and every set of formula Γ , we have that $\Gamma \vdash_{\Sigma} A$ implies $\Gamma \models A$.

Completeness

A proof system Σ is **complete** if for every formula A and every set of formula Γ , we have that $\Gamma \models A$ implies $\Gamma \vdash_{\Sigma} A$.

In other words,

- a sound Σ derives only logically implied formulas, and
- a complete Σ derives all logically implied formulas.

The Hilbert proof system is sound and complete

Theorem (Soundness of Hilbert proof system)

If $\Gamma \vdash_{\mathcal{H}} A$ then $\Gamma \models A$.

Theorem (Completeness of Hilbert proof system)

If $\Gamma \models A$ then $\Gamma \vdash_{\mathcal{H}} A$.

Exercise

Prove that $\mathcal{H}$ is a sound proof system.

Completeness is much more difficult to prove ...

Importance of the Hilbert proof system

The main objective of Hilbert was to find the smallest set of axioms and inference rules from which it was possible to derive all the tautologies.

However, proofs and inferences in the Hilbert proof system are not very intuitive. Other proof systems are more intuitive (see later). As a matter of facts, nobody is practically using Hilbert system for deduction.

Why is it so important then?

The Hilbert proof systems was the **first** formal proof system formally defined and studied for any logic. At that time, proof systems were the only hope for mechanizing reasoning!